

GOKU TEAM · AGENTIC AI BUILD WEEK 2026

KFC Ordering Agent

KFC customers are already messaging on Messenger. We turn the conversation into a completed order — with an AI agent that plans, acts through tools, and verifies itself.

F8B · AI Engineering · Product · Problem 4 — Conversational ordering

01 · PROBLEM INSIGHT

Customers are already in chat. The operator is the ceiling.

WHO

High-intent customers who message the fanpage before opening the app.

GOAL

Finish the order inside the chat, no channel switch.

FRICTION

Ordering means leaving chat for the app — conversion leaks midway.

ROOT CAUSE

100% handled by hand — the operator's speed caps chat revenue.

EVIDENCE

KFC brief: 250+ stores, millions of customers a month, no conversational ordering.

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ONE INSIGHT

KFC's chat channel isn't short on customers — it's short on someone to close orders 24/7.

The agent doesn't create new demand. It harvests demand that leaks away every day.

02 · AGENTIC WORKFLOW

Goal → Plan → Tools → Act → Verify, on one real message

Customer: "deliver to 350 Tran Hung Dao, D5, phone 090..., pay cash please"

	→ PLAN	→ TOOLS	→ ACT	→ VERIFY
GOAL				
Understand — close a	Resolve store →	set_delivery_info ·	Route to KFC	State machine +
COD order to a District	check sold-out items	select_payment_method	Nguyen Trai (open to	service validate, two
5 address.	→ close COD.	(2 of 14 tools).	22:00) · create order	layers · wrong step =
			KFC-0001.	tool locked.

Not a black box — the staff console shows the decision log each turn. And tool discipline: 14 tools locked across 11 order states, so the payment tool doesn't exist to the LLM until an order is confirmed.

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03 · WHY IT WINS

The LLM holds the conversation. The state machine holds the money.

Hybrid agent + state machine

The LLM never decides price or order on its own — every number comes from the DB through tools, validated twice.

→ Orders close in under 2 minutes, never a made-up price.

Store-aware, 250 stores

Orders route to the right store by address, knowing opening hours and sold-out items before accepting.

→ Never accepts an order it can't deliver — operations you can trust.

Humans in the right place

Complaints and sensitive cases hand off; staff take over in one tap, and the customer always knows who they're talking to.

→ Controlled automation — enterprise-grade.

Why can't a general assistant do this? Because it isn't allowed to handle money. This system is — safely.

04 · EVIDENCE + IMPACT

What's measured, what's assumed — labeled honestly

Level 2 · Benchmark — measured	Level 1 · Assumption — labeled	Impact at scale
24/24 Vietnamese NLU auto-eval with trap cases against fabricated items · re-run after every change (caught 2 regressions in a day).	+2.7% AOV uplift, backtested on 4,964 simulated POS transactions at 20% acceptance — conservative vs. KFC's 10–15% brief estimate.	Chat orders/mo × AOV × uplift + recovered chat conversion (now 0, hand-handled) + reduced operator load · pilot measures via A/B.

11-step	24/24	Threshold	Auto
Order completion flow, running live	NLU accuracy · gpt-5.4	Upsell: "+11k → free ship, save 20k"	Best voucher applied, nothing to memorize

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05 · DEMO + CLOSE

60-second demo — ending on the customer's outcome

0-10S · GOAL	→ 10-20S · TRIGGER	→ 20-45S · AGENT ACTS	→ 45-55S · OUTCOME	→ 55-60S · PROOF
Hungry customer messages the fanpage like texting a friend.	"Get me a fried-chicken combo + Pepsi."	Tool-trace runs: cart → voucher-threshold upsell → store → COD.	Order code KFC-xxxx + tracking link back to the customer.	Admin: funnel + AOV + NLU 24/24 tick live.

From 100% hand-answered → a sales agent that works 24/7. Pilotable today on KFC's real fanpage.

Dual screen: the phone shows the customer experience, the staff console shows the agent's brain via tool-trace — judges see both sides of one conversation.

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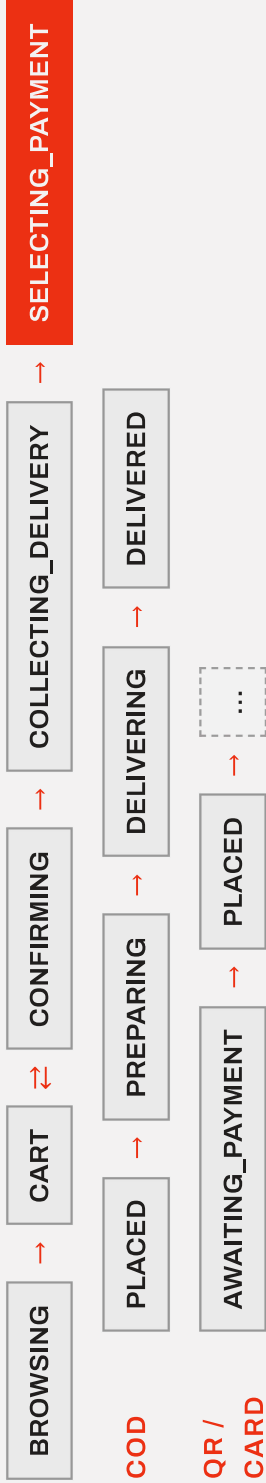
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Appendix · A1–A8

Evidence for Q&A. We answer first and show an appendix slide only when a claim needs support.

A1 · AGENTIC WORKFLOW

An 11-state machine drives the order lifecycle



Tools locked by state	Two-way sync	Many customers at once
Each state opens only a subset of the 14 tools — the LLM can't check out before an order is confirmed, even under prompt injection.	Staff change the order state → the agent session syncs → the customer gets a push; a drifted session self-heals on the next turn.	Per-conversation lock (CAS) + queue + dedupe by mid + consecutive messages merged into one LLM turn.

A2 · TECHNICAL ARCHITECTURE

Architecture is evidence — not the main character

MESSENGER → **WEBHOOK** · **VERCEL** → **AGENT** · **GPT-5.4** → **SERVICE LAYER** · **TS** → **NEON POSTGRES**
Customers message Verify HMAC signature Vercel AI SDK · 14 Order state machine · All state — serverless,
the real fanpage (Meta · dedupe mid · lock state-gated tools · store · voucher · 100% stateless app.
dev mode). / queue. retry + graceful. loyalty · upsell.

Portable by design

The core is framework-free TypeScript —
containerize onto KFC's standard infra
(Azure) in ~1 day, swapping only a thin
route layer.

Meta-policy correct

Push outside the 24h window uses
POST_PURCHASE_UPDATE; marketing
push goes via Recurring Notifications opt-
in (roadmap).

Upsell without the LLM

Suggestions = daypart + co-
occurrence counted from 4,964 POS
transactions + voucher thresholds —
deterministic, explainable.

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A3 · SAFETY + HUMAN OVERSIGHT

Automation with brakes

Money never touches the LLM

Price, total and voucher are computed only by the service from the DB. Tools validate an item exists before adding to cart. Saying "ordered" requires a real order code from a tool.

Customer data

Phone numbers masked in admin and logs · webhook verifies signatures and rejects unsigned requests in production · secrets in env · public privacy policy.

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Human oversight

Complaints and sensitive cases (allergy, refund) hand off. Staff take over in one tap — the bot goes silent instantly, and the customer is told clearly whether they're talking to a person or a bot.

Fault tolerance

LLM error → retry, then apologize gracefully; messages are never lost (peek / mark-after) · idempotent payments · the /pay page blocks already-processed orders.

A4 · USER RESEARCH

Tested for real, found real bugs, fixed same-day

OBSERVED After delivery, the customer's next order is rejected.	OBSERVED Bot says "I'll order now" but creates no order.	OBSERVED Bot makes up a combo price from old chat history.
ROOT CAUSE Session stuck in the old state + cart not cleared after the order.	ROOT CAUSE LLM "says but doesn't do" — missing a say=do constraint.	ROOT CAUSE In a stuck state the agent loses tools, so it acts in words.
FIX + VERIFY Self-healing session + two-way state sync + cart clear — smoke test 6/6.	FIX + VERIFY SAY=DO rule + absolute priority for the order tool — reproducing eval case passes.	FIX + VERIFY TOOL > HISTORY in the prompt + enough tools open in every valid state.

Every bug becomes a case in the eval suite — the system never repeats an old error.

A5 · IMPACT / ROI

Formula first, numbers second

Value / year = chat orders per month × AOV × uplift × 12 + recovered chat conversion + page-operator hours

Measured (mock, transparent)	From the KFC brief	Pilot measures for real
Backtest of 4,964 POS transactions: AOV +2.7% at 20% acceptance · voucher-threshold upsell lifts basket value right in the demo.	KFC estimates context-right upsell lifts AOV 10–15% · the chat channel currently converts ≈0 because 100% is hand-handled.	A/B on the real fanpage: completion rate, AOV delta, response time, CSAT — 2 weeks of real numbers replace every assumption.

LLM cost is under 2% of order value — and only the chat channel uses the LLM; the upsell engine runs deterministically, for free.

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A6 · ALTERNATIVES

Compared with two familiar paths

Rule-based bot (button menus)

Can't parse "2 spicy chickens, mild, deliver D5, pay cash" in one sentence · every flow change means recoding the button tree.

✗ **RIGID, UNNATURAL**

General LLM assistant

Answers well but isn't allowed to handle money: no state machine, no tool gating
→ made-up prices and mis-orders are a certainty.

✗ **UNCONTROLLED**

Our hybrid

The LLM handles natural language · a state machine + 14 step-locked tools handle money and orders · handoff handles exceptions.

✓ **NATURAL + SAFE**

GenAI where language is needed — deterministic where money is touched.

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A7 · EVALUATION RESULTS

Automated eval — we don't trust vibes

SUITE	CASES	WHAT IT CHECKS
NLU	9	Natural item ordering, quantity, ambiguous size → re-ask, multiple intents in one sentence
Flow	8	confirm → delivery → COD / QR → tracking → cancel, all by the rules
Guardrail	4	nonexistent item → no fabrication · early COD mention → still creates a real order
Store-aware	3	resolve the right district · store closed → switch · sold out → suggest an alternative

Run through the real agent + real services + real DB — **24/24 · gpt-5.4**. Today it once dropped to 21/24 after a change — caught and fixed before deploy.

A8 · ROADMAP + LIMITATIONS

Knowing exactly where we stand

Current limits (demo)

- Synthetic data — the brief confirmed the direction is right
absent real data
- Transparent mock payments — no real gateway connected yet
- Stores matched by district, no delivery-radius check yet
- Messenger dev mode — users need a tester role

Roadmap after pilot

- Connect KFC's real POS / OMS + loyalty (schema already mirrored)
- VNPay / MoMo webhook · distance API · per-store menu overlay
- Zalo on the same agent core (just add an adapter) · P2 kiosk on the same upsell engine
- Recurring Notifications opt-in · anti-fraud (COD limit, blacklist)

Q&A PREP · INTERNAL

Q&A — Answer → Support → Connect

1 · ANSWER

Conclusion first, no hedging.
"Yes — the agent handles many customers at once."

→ 2 · SUPPORT

Exactly one fact.
"Per-conversation lock + queue, tested with two phones in parallel."

→ 3 · CONNECT

Back to user value.
"So at the lunch peak, no customer waits."

Asked something we don't have: "Not validated yet — what we know is X, and we'll test it with Y in the pilot." · Opening Q&A:
"May I gather the questions first?" (0:20 gather · 1:20 answer).

Q&A PREP · INTERNAL

Six one-line answers, one per rubric criterion

Agentic AI	The agent plans and acts through 14 state-locked tools — the tool-trace shows each decision in real time.
Track fit	Solves KFC's P4 exactly: the chat channel is 100% hand-handled; we turn it into the first automated sales channel.
Execution	An 11-step flow live on Messenger: order → pay → track → deliver — eval 24/24, smoke 25/25.
Impact	Orders close in chat in under 2 minutes, voucher-threshold upsell lifts AOV, the page team is freed from repetitive work.
Creativity	Hybrid: the LLM holds the conversation, the state machine holds the money — a general assistant can't handle money; this system can, safely.
Clarity	One sentence to retell: "Text KFC like texting a friend — and the order runs itself to your door."